When Interpretability Hurts Information Retrieval

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Abstract

Performance in Information Retrieval (IR) heavily depends on the quality of embeddings. While dense embeddings excel at capturing complex patterns, their lack of interpretability limits user trust and control. Sparse representations, such as those from Sparse Autoencoders (SAEs), offer greater transparency by isolating distinct concepts but may sacrifice retrieval accuracy. In this study, we compare multimodal (CLIP) and sparse (SAE) embeddings in image-based IR against more traditional, dense and unimodal representations, evaluating both accuracy and computational efficiency. Our experiments reveal that interpretable sparse and multimodal representations reduce the IR performance.

1. Introduction

Image-based Information Retrieval (IR) systems rely on dense visual embeddings to identify relevant results. While highly effective at capturing complex patterns, these dense representations are not easily interpretable, offering little insight into why particular items are retrieved - a limitation in domains where explainability and user trust are critical.

Recent advances in representation learning have introduced sparse and multimodal alternatives to address these challenges. Sparse representations, such as those generated by SAEs (Gao et al., 2024), decompose complex data into human-understandable concepts. This interpretability enables users to identify activated features and even control results by adjusting concept weights. Similarly, multimodal models like CLIP (Radford et al., 2021) jointly train image and text encoders, allowing IR systems to support natural language queries by combining visual and textual information.

Despite these advantages, the adoption of sparse and multimodal representations raises a question: Does improved interpretability and flexibility come at the expense of retrieval effectiveness? While dense embeddings have been shown to perform well in IR tasks, sparse and multimodal approaches — although promising in terms of transparency — may result in reduced accuracy. Prior work such as (Rao et al., 2024) demonstrates that SAEs can uncover meaningful and interpretable structure in CLIP features, focusing primarily on classification tasks.

In this project, we shift the focus to retrieval, investigating how such interpretable representations perform in image-based IR scenarios compared to dense baselines, hypothesizing that their transparency and controllability may come at the cost of retrieval quality.

Our contributions are summarized as follows:

- We investigate the impact of interpretability and multimodality in visual representations on image retrieval performance, examining multimodal CLIP embeddings and sparse embeddings from SAEs.
- We evaluate these methods on several datasets measuring retrieval quality using IR specific metrics.
- We examine how top-K selection in sparse embeddings impacts the relevance and interpretability of retrieval outcomes.
- We highlight the trade-off between interpretability and retrieval effectiveness, showing that greater transparency may come at the cost retrival quality.

2. Methodology

In our experiments, we focused on evaluating the impact of sparse (SAE) and multimodal (CLIP) representations on IR. Our main goal was to assess whether the interpretability of representations leads to reduced retrieval quality.

We used the Image Similarity Challenge (ISC 2021) dataset as the primary benchmark, along with CIFAR-10 and STL-10 for smaller-scale demonstrations and testing. The ISC dataset consists of 50,000 query images and 1 million reference images. Of these, 10,000 query images are modified versions of the reference images. These adjustments included rotations, screenshot overlays, horizontal flips, color scaling, blurring, changes in brightness, and similar transformations. The goal was to identify the matching reference image or to determine that no match exists.

For sparse representations, we used code and pretrained checkpoints from the official Discover-then-Name repository (Müller et al., 2024).

We implemented a codebase for extracting CLIP embeddings, loading SAE checkpoints, generating sparse representations, and performing IR using cosine similarity (for CLIP) and L1 distance (for SAE).

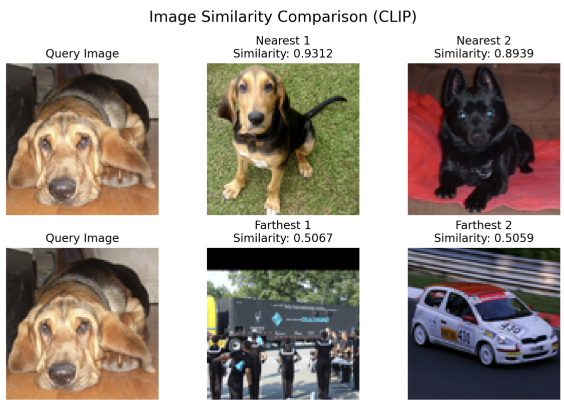
Evaluation included:

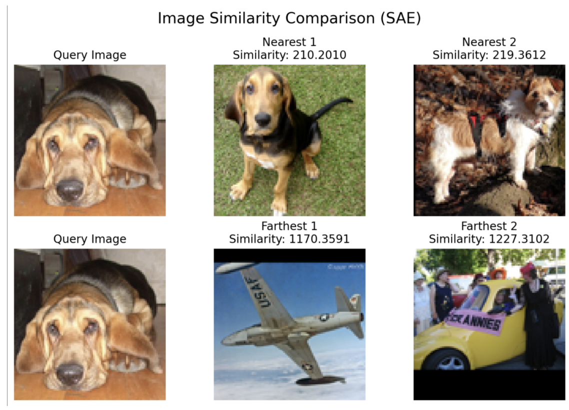
- computing metrics (Douze et al., 2022):
  - **precision** and **precision-top-K**, where true positive means that the ground truth is present in the top K retrieved neighbours (correct match). Precision is computed for a given similarity **threshold**, which determines whether a nearest neighbour is considered a match or not;
  - **recall** and **recall-top-K**, measuring whether the relevant item is present within the top K neighbours. Similarly, recall depends on the selected **threshold** value;
  - μ**AP** (micro average precision), computed as the area under the precision-recall curve obtained by sorting retrieved pairs by similarity scores.
- boxplots and histograms of similarity for correct and incorrect matches;
- experiments with top-K active SAE features.

The full codebase and experiments are available in our repository.

3. Experiments

First, we examined the nearest neighbours determined by both SAE and CLIP. The experiment was conducted on a 250-image sample from the STL-10 dataset. The results are visualized in Figure 1.

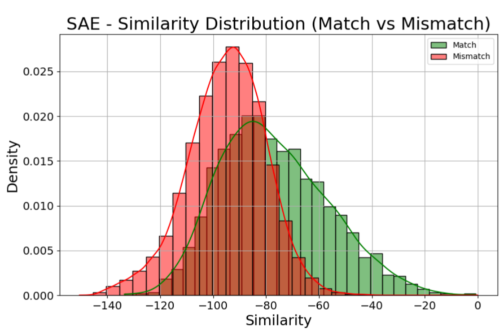


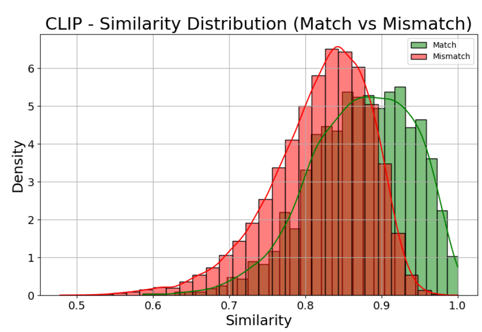


(a) (b)

Figure 1: **Comparing neighbourhoods in CLIP and SAE.** We show an example from STL-10 dataset with closest and farthest images by CLIP and SAE similarities. We find that both approaches yield similar results despite different representations.

For each query image of ISC dataset, the closest neighbours were identified by analyzing similarity between their representations. A match was accepted if similarity threshold was met. Setting a threshold on the confidence amounts to taking a hard decision whether there is a detection or not. Calculations were run for a range of threshold values. The evaluation metric used to assess the results was μAP, which reached a level of 0.0647 for SAE — ten times lower than the baseline from the ISC 2021 (Douze et al., 2022). For CLIP, the result was slightly lower at 0.0399 (see Table 1). Dense representations (512-dimensional vectors) required substantially less computational power compared to their sparse counterparts (4096-dimensional vectors).





(a) (b)

Figure 2: **Similarity scores distribution** of matched and mismatched reference images, illustrating the difference in similarity values (cosine similarity for CLIP and negative L1 distance for SAE) between correct and incorrect retrievals.

We analyzed the difference in similarity scores between correctly matched and unmatched images for both methods (see Figure 2). It was observed that correctly assigned references are closer to the query image than incorrect matches. This suggests that despite the lower overall retrieval quality, both methods are still capable of retrieving relevant content.

	CLIP	**SAE**
Recall	0.075	0.075
Precision	0.1047	**0.3004**
μAP	0.0399	**0.0647**
RAM usage [GB]	**1.6**	3.3
Calculation time [s]	**37**	1087

Table 1: **Results of neighbours search process.** While Recall, Precision and μAP were calculated on whole query images dataset (50 000 instances), RAM usage and process time were examined on subset of first 500 query images. The proportions are preserved, as 104 of selected images had Ground Truth reference (approx. 20% of dataset). One can set the similarity threshold and therefore adjust the tradeoff between Precision and Recall. We have chosen a similarity threshold of 0.7507 for CLIP and -69.96 for SAE, so that Recall reaches 0.075 and Precision may be compared.

Subsequently, SAE representations were modified to retain only the top-K most active neurons, where $K \in \{5, 10, 20\}$; that is, only the K highest activation values were preserved, while the others were set to zero. Nearest neighbours were then found using these modified representations. From the examples in Figure 3, we can conclude that selecting top-K activations influences the choice of nearest neighbours. However, the semantic content of the closest images remains similar — for instance, in Figure 3, all neighbouring images depict forests, similarly to the closest original matches from both CLIP and SAE representations.

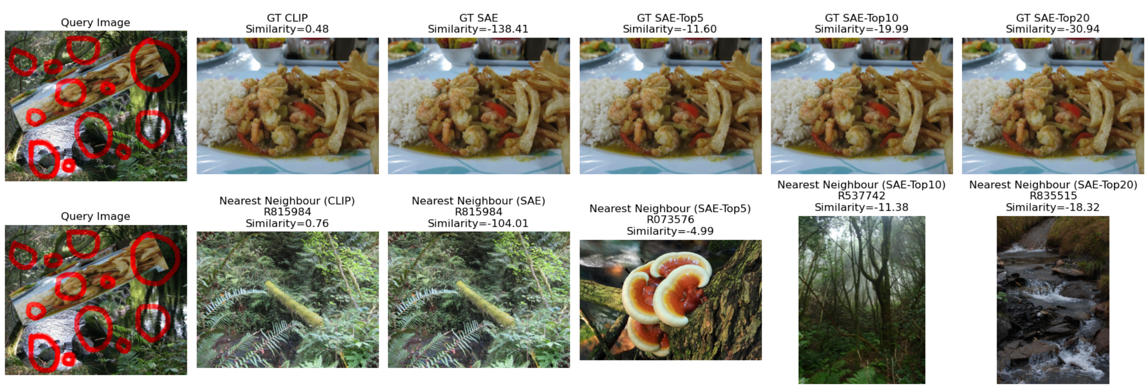


Figure 3: **Nearest neighbours with top-K method.** The original photo (depicting lunch on a plate) was resized, rotated, pasted on forest background and red circles were drawn. It is beyond doubt that highest activation values were the concepts of forest or red polypore-like mushroom.

4. Conclusions

Our experiments showed that while sparse and multimodal representations offer improved interpretability, they also reduce the IR performance on images.

Future work should explore applying these techniques to text data to determine whether the observed reduction in retrieval performance also extends to textual input.

References

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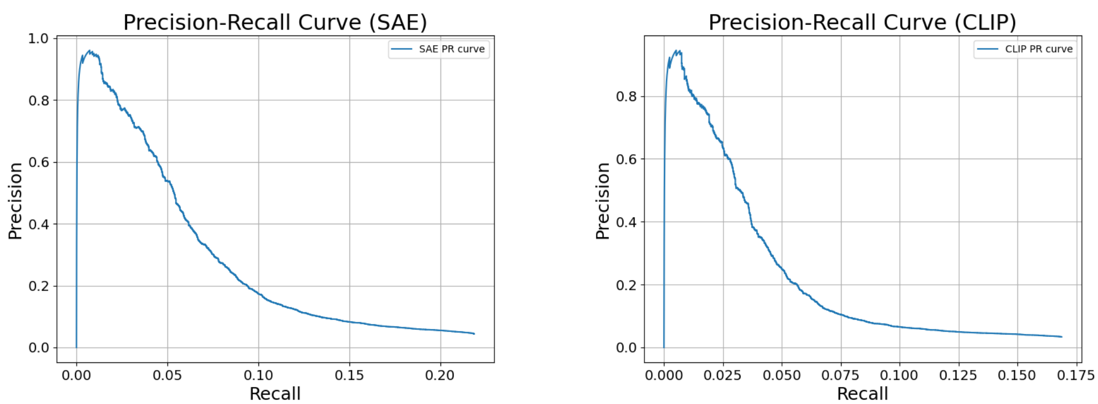
5. Appendix



(a) (b)

Figure 4: **Precision-Recall curves**.

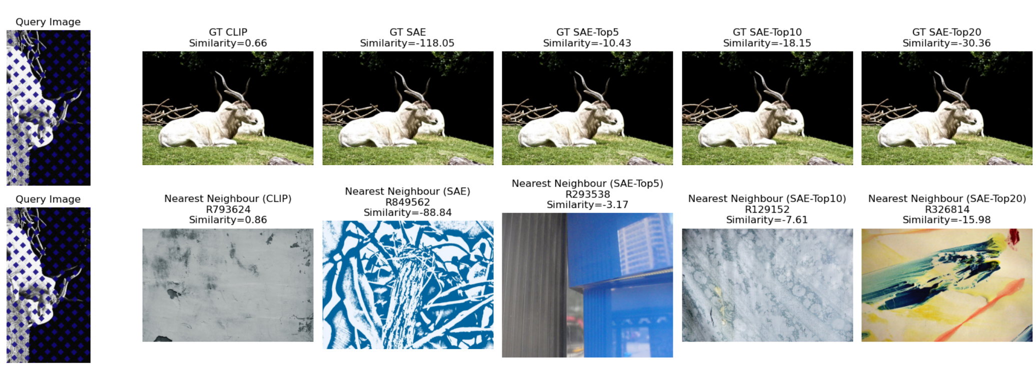


Figure 5: **Nearest neighbours with top-K method.** The original photo (depicting horny animal) was resized, rotated, grayscaled and blue diamonds were drawn. It is understandable that highest activation values were the concepts of color blue.